VIDYA BHARTI PRATIBHA KHOJ AND PRATIBHA CHAYAN OMR SCANNER

VERSION 1.0.0

USER & BUILD DOCUMENTATION

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1. INTRODUCTION

VB OMR Scanner is an automated Optical Mark Recognition (OMR) application designed specifically for processing Vidya Bharti Pratibha Khoj and Pratibha Chayan examination answer sheets.

Features:

- Batch processing of scanned OMR sheets

- Automatic detection of:

- Candidate Name (25 characters)
- Roll Number (3 digits)
- Class (3-10)
- Section (A-H)
- School Code (4 digits)
- 60 Multiple Choice Questions (A, B, C, D)
- Configurable scoring system
- Excel export functionality
- Results ranking and statistics

2. SYSTEM REQUIREMENTS

MINIMUM REQUIREMENTS:

- Operating System: macOS 10.13+ or Windows 10+
- RAM: 4 GB (8 GB recommended)
- Storage: 500 MB free space
- Display: 1280 x 720 resolution

FOR BUILDING FROM SOURCE:

- Python 3.9 or higher
- pip (Python package manager)
- 2 GB additional storage for build dependencies

3. BUILDING THE APPLICATION

3.1 PREREQUISITES

1. Install Python 3.9 or higher from https://python.org

2. Clone or download the project source code

3. Open Terminal (macOS) or Command Prompt (Windows)

4. Navigate to the project directory:

cd /path/to/omrsheetproj

5. Create and activate a virtual environment:

macOS/Linux:

---

python3 -m venv venv
source venv/bin/activate

Windows:

---

python -m venv venv
venv\Scripts\activate

6. Install dependencies:

pip install --upgrade pip
pip install opencv-python numpy pywebview openpyxl Pillow jsonschema pyinstaller

3.2 BUILDING FOR macOS (.app / .dmg)

METHOD 1: Using Build Script (Recommended)

1. Make the build script executable:

chmod +x scripts/build_mac.sh

2. Run the build script:

./scripts/build_mac.sh

3. The application will be created at:

dist/VB OMR Scanner.app

METHOD 2: Manual Build

1. Activate the virtual environment:

source venv/bin/activate

2. Run PyInstaller:

pyinstaller build.spec --clean

3. The application will be created at:

dist/VB OMR Scanner.app

CREATING A DMG FILE:

1. After building the .app, create a DMG:

hdiutil create -volname "VB OMR Scanner" \

-srcfolder "dist/VB OMR Scanner.app" \
-ov -format UDZO \
"dist/VB_OMR_Scanner.dmg"

2. The DMG file will be created at:

dist/VB_OMR_Scanner.dmg

3.3 BUILDING FOR WINDOWS (.exe)

METHOD 1: Using Build Script (Recommended)

1. Open Command Prompt as Administrator

2. Navigate to the project directory:

cd C:\path\to\omrsheetproj

3. Run the build script:

scripts\build_windows.bat

4. The executable will be created at:

dist\VB OMR Scanner.exe

METHOD 2: Manual Build

1. Activate the virtual environment:

venv\Scripts\activate

2. Run PyInstaller:

pyinstaller build.spec --clean

3. The executable will be created at:

dist\VB OMR Scanner.exe

4. USING THE APPLICATION

4.1 GETTING STARTED

1. Launch the application:

- macOS: Double-click "VB OMR Scanner.app"
- Windows: Double-click "VB OMR Scanner.exe"

2. The application will open with the Dashboard view

3. Before processing sheets, you must set up an Answer Key

4.2 DASHBOARD

The Dashboard provides quick access to main features:

- PROCESS NEW BATCH: Start processing OMR sheets
- VIEW RESULTS: View processed results and statistics
- ANSWER KEY: Configure the correct answers for scoring

Quick Start Guide:

1. Set up your Answer Key (required for scoring)

2. Select a folder containing scanned OMR sheets

3. Start processing and monitor progress

4. Export results to Excel

4.3 SETTING UP ANSWER KEY

1. Click "Answer Key" in the sidebar or dashboard

2. For each question (1-60), select the correct answer (A, B, C, or D)

- Click on the option button to select it
- The selected answer will be highlighted

3. The counter shows how many questions have been answered

(e.g., "Filled: 45 / 60")

4. Click "Save Answer Key" to save your configuration

IMPORTING ANSWER KEY FROM FILE:

1. Click "Import from File"

2. Select a text file with answers in format:

1,A

2,B
3,C
... (one question per line)

4.4 PROCESSING OMR SHEETS

1. Click "Process Sheets" in the sidebar

2. INPUT FOLDER:

- Click "Browse" next to Input Folder
- Select the folder containing scanned OMR sheet images
- Supported formats: JPG, JPEG, PNG, TIFF, BMP
- The file count will be displayed

3. OUTPUT FILE (Optional):

- Click "Browse" next to Output File
- Choose where to save the Excel results
- If not specified, results will be saved in the input folder

4. SCORING:

- Check "Enable Scoring" to calculate marks (requires answer key)
- Uncheck to only extract data without scoring

5. Click "START PROCESSING"

6. Monitor progress:

- Progress bar shows completion percentage
- Current file being processed
- Processed count (e.g., "15 / 50")
- Elapsed time and estimated remaining time

7. Click "Cancel" to stop processing at any time

RECOMMENDED SCAN SETTINGS:

- Resolution: 300 DPI
- Color: Grayscale or Color
- Format: JPEG or PNG
- Orientation: Portrait (correct side up)

4.5 VIEWING RESULTS

1. Click "Results" in the sidebar

2. STATISTICS BAR shows:

- Total: Number of processed sheets
- Avg Score: Average score across all sheets
- Highest: Highest score achieved
- Lowest: Lowest score achieved

3. RESULTS TABLE displays:

- Rank: Position based on score
- Name: Detected candidate name
- Roll No.: Detected roll number
- Class: Detected class (3-10)
- Section: Detected section (A-H)
- School Code: Detected school code
- Correct: Number of correct answers
- Wrong: Number of wrong answers
- Score: Calculated score
- %: Percentage score

4. Click on any row to view detailed information about that candidate

4.6 EXPORTING RESULTS

1. Go to "Results" page

2. Click "Export to Excel" button

3. Choose the save location and filename

4. The Excel file will contain:

- Sheet 1: Complete results with all data
- All candidate information
- Individual question responses
- Scores and rankings

4.7 SETTINGS

1. Click "Settings" in the sidebar

2. SCORING CONFIGURATION:

• Correct Answer Marks: Points for each correct answer
Default: 1.0

• Wrong Answer Marks: Points deducted for wrong answers
Default: -0.25 (negative marking)

• Unanswered Marks: Points for unanswered questions
Default: 0

3. Click "Save Settings" to apply changes

5. OMR SHEET SPECIFICATIONS

The application is configured for Vidya Bharti examination OMR sheets with:

PAGE DIMENSIONS:

• Width: 2480 pixels (at 300 DPI)
• Height: 3507 pixels (at 300 DPI)

FIELDS DETECTED:

1. CANDIDATE NAME

• Location: Top-left area
• Format: 25 columns x 26 rows (A-Z)
• Bubble size: 20 x 20 pixels

2. ROLL NUMBER

• Location: Top-right area
• Format: 3 digits (0-9 each)
• Direction: Vertical bubbles

3. CLASS

• Location: Top-right area (after roll number)
• Options: 3, 4, 5, 6, 7, 8, 9, 10
• Direction: Vertical bubbles

4. SECTION

• Location: Top-right area (after class)
• Options: A, B, C, D, E, F, G, H

- Direction: Vertical bubbles

5. SCHOOL CODE

- Location: Top-right area (after section)
- Format: 4 digits (0-9 each)
- Direction: Vertical bubbles

6. ANSWER SECTION

- Location: Bottom half of sheet
- Questions: 60 (arranged in 6 columns of 10)
- Options: A, B, C, D per question
- Direction: Horizontal bubbles

6. TROUBLESHOOTING

ISSUE: Application won't start

SOLUTION:

- macOS: Right-click and select "Open" if blocked by Gatekeeper
- Windows: Run as Administrator if permission errors occur

ISSUE: Poor detection accuracy

SOLUTIONS:

- Ensure scans are at 300 DPI resolution
- Sheets should be properly aligned (not skewed)
- Use good quality scans without shadows or artifacts
- Ensure bubbles are filled darkly and completely

ISSUE: "No answer key loaded" warning

SOLUTION:

- Go to Answer Key page and configure all 60 answers
- Click "Save Answer Key" before processing

ISSUE: Some questions show as "MULTI"

EXPLANATION:

- Multiple bubbles were detected as filled for that question
- This may indicate the candidate marked multiple answers
- Check the original sheet to verify

ISSUE: Some questions show as empty

POSSIBLE CAUSES:

- Bubble was not filled darkly enough
- Bubble was filled outside the designated area
- Scan quality issue in that area

ISSUE: Wrong class detected

EXPLANATION:

- Class uses values 3-10 (not 0-9)
- Ensure the correct row is marked on the sheet

ISSUE: Build fails with missing modules

SOLUTION:

- Ensure all dependencies are installed:

pip install opencv-python numpy pywebview openpyxl Pillow jsonschema pyinstaller

- Try: pip install --upgrade --force-reinstall [package-name]

7. SUPPORT

For technical support or bug reports:

- Check the troubleshooting section above
- Ensure you are using the latest version
- Provide details about your system and the issue

When reporting issues, please include:

- Operating system and version
- Application version
- Steps to reproduce the issue
- Sample images (if applicable, with personal data removed)

END OF DOCUMENTATION

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